$-\Delta_x u = 0$ (Laplace's eqn)

$-\Delta_x u = f$ (Poisson's eqn)

$\vec{x} \in \mathbb{R}^n$

"in"

Last time: Solution of $-\Delta u = f$ via a fundamental solution $\Phi(x)$

$\alpha(n)$ = S.A. of n-sphere

$$\Phi(x) = \begin{cases} \dfrac{1}{n(n-2)\alpha(n)} \dfrac{1}{|x|^{n-2}}, & n \geq 3 \\ -\dfrac{1}{2\pi} \ln(|x|), & n = 2 \end{cases}$$

$|x| = \sqrt{x_1^2 + \cdots + x_n^2}$

Claim: $u(x) = \int_{\mathbb{R}^n} f(y)\, \Phi(x-y)\, dy$ solves $\boxed{-\Delta u = f \text{ in } \mathbb{R}^n}$

Convolution

If f, g are defined on $\mathbb{R}^n$, the <u>convolution</u> $f * g$ is another function defined by

$$(f*g)(x) = \int_{\mathbb{R}^n} f(x-y)\,g(y)\,dy = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} f(x-y)\,g(y)\,dy_1\,dy_2\cdots dy_n$$

$n=1$ example:

$f(x) = x$

$$g(x) = \begin{cases} x^2, & |x| \le 10 \\ 0, & |x| > 10 \end{cases}$$

$$(f*g)(x) = \int_{-\infty}^{\infty} f(x-y)\,g(y)\,dy = \int_{-10}^{10} (x-y)\,y^2\,dy + \int_{10}^{\infty} (x-y)\,0\,dy + \int_{-\infty}^{-10} (x-y)\,0\,dy$$

$$\cdots = \int_{-10}^{10} (x-y)y^2\,dy = \int_{-10}^{10} (xy^2 - y^3)\,dy = x\frac{y^3}{3}\Big|_{-10}^{10} - \frac{y^4}{4}\Big|_{-10}^{10}$$

$$= \frac{x}{3}(1{,}000 + 1{,}000) + 0$$

$$= \frac{2{,}000}{3}x$$

Two facts about convolution:

1) $f * g = g * f$

2) If f, g are both bounded and smooth, then $(f * g)_{x_i} = f_{x_i} * g = f * (g_{x_i})$

Thm If $f \in C_c^2(\mathbb{R}^n)$, then

$$u(x) = (\Phi * f)(x) = \int_{\mathbb{R}^n} \Phi(x-y) f(y)\, dy$$

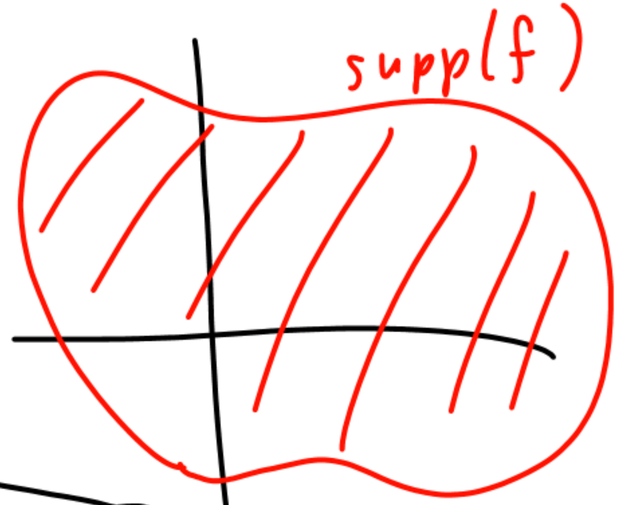
is $C^2(\mathbb{R}^n)$ and solves $-\Delta u = f$ in $\mathbb{R}^n$.

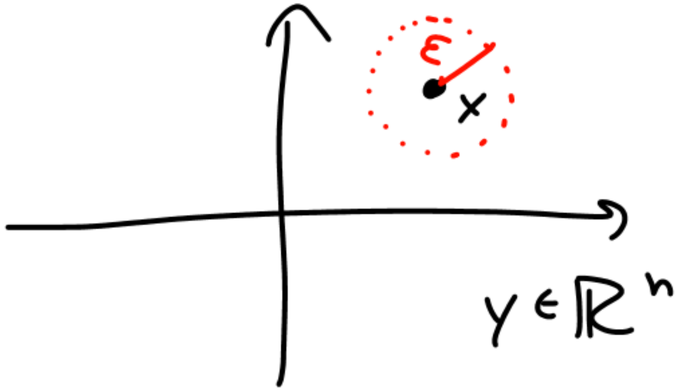
Pf See Evans for proof that $u \in C^2(\mathbb{R}^n)$.

To prove $-\Delta u = f$, recall

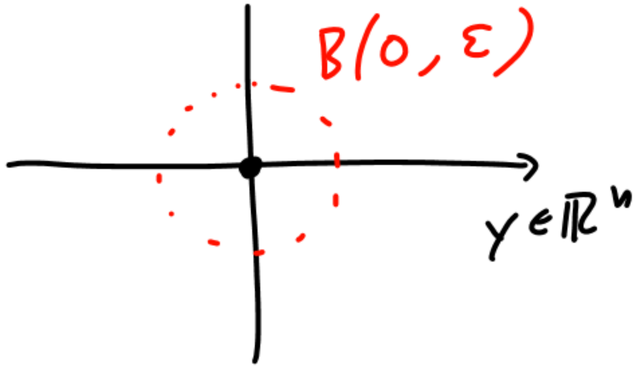
$\Delta_x \Phi(x-y) = 0$ when $x \neq y$.

Q: Does $\Delta_x u = \Delta_x \int \Phi(x-y) f(y)\, dy = \int \Delta_x \Phi(x-y) f(y)\, dy = 0$? No.





or equivalently,



$$u(x) = \int_{\mathbb{R}^n} \Phi(x-y) f(y)\,dy = \int_{\mathbb{R}^n} \Phi(y) f(x-y)\,dy$$

$$= \int_{B(0,\varepsilon)} \Phi(y) f(x-y)\,dy + \int_{\mathbb{R}^n \setminus B(0,\varepsilon)} \Phi(y) f(x-y)\,dy$$

$$\Delta u(x) = \int_{B(0,\varepsilon)} \Phi(y) \Delta_x f(x-y)\,dy + \int_{\mathbb{R}^n \setminus B(0,\varepsilon)} \Phi(y) \Delta_x f(x-y)\,dy$$

$$= I_\varepsilon + J_\varepsilon$$

$B(0,\varepsilon) = \{y \in \mathbb{R}^n : |y| < \varepsilon\}$

$\mathbb{R}^n \setminus B(0,\varepsilon) =$

$= \{y \in \mathbb{R}^n : |y| \geq \varepsilon\}$

$$I_\varepsilon = \int_{B(0,\varepsilon)} \Delta_x f(x-y)\Phi(y)\,dy$$

By assumption, $f \in C_c^2(\mathbb{R}^n)$.

$\Rightarrow \Delta_x f = \sum_{i=1}^{n} f_{x_i x_i}$ is continuous in a compact set

$\Rightarrow \Delta_x f$ is bounded, $|\Delta_x f(x)| \le K_f$ for all x.

$$|I_\varepsilon| = \left| \int_{B(0,\varepsilon)} \Delta_x f(x-y)\Phi(y)\,dy \right| \le \int_{B(0,\varepsilon)} |\Delta_x f(x-y)||\Phi(y)|\,dy$$

$$\le \int_{B(0,\varepsilon)} K_f |\Phi(y)|\,dy = K_f \int_{B(0,\varepsilon)} |\Phi(y)|\,dy$$

$$|I_\varepsilon| \le K_f \int_{B(0,\varepsilon)} |\Phi(y)|\,dy \qquad n \ge 3$$

$$= K_f \int_{B(0,\varepsilon)} c_n \frac{1}{|y|^{n-2}}\,dy$$

$\partial B(0,r) = \{y \in \mathbb{R}^n : |y| = r\}$

(Recall: in $n=3$ case,

$$\int_{B(0,\varepsilon)} c_n \frac{1}{|y|^{n-2}}\,dy = \int_0^\varepsilon \int_0^\pi \int_0^{2\pi} c_n \frac{1}{\rho} \rho^2 \sin\phi \, d\phi \, d\theta \, d\rho$$

$$= c_n \int_0^\varepsilon \rho \, d\rho \int_0^\pi \sin\phi \, d\phi \cdot 2\pi = C \varepsilon^2 .)$$

In general, we always have

$$\int_{B(0,\varepsilon)} c_n \frac{1}{|y|^{n-2}}\,dy = \int_0^\varepsilon \int_{\partial B(0,\rho)} \frac{1}{\rho^{n-2}}\,dS(\omega)\,d\rho = c_n \int_0^\varepsilon \frac{\rho^{n-1}}{\rho^{n-2}}\,d\rho = c_n \varepsilon^2$$

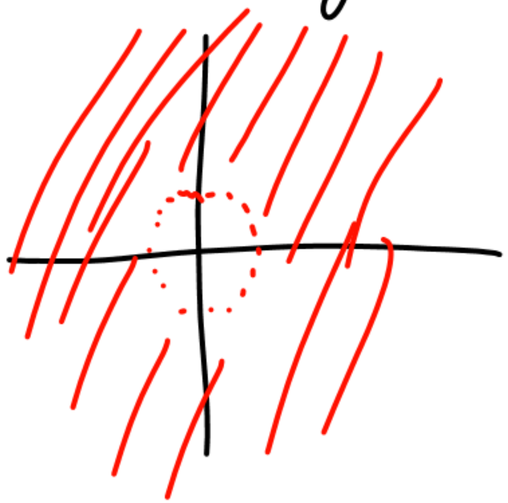
$$J_\varepsilon = \int_{\mathbb{R}^n \setminus B(0,\varepsilon)} \Phi(y) \Delta_x f(x-y)\, dy$$

$$\boxed{\int_U \operatorname{div}(F)\, dy = \int_{\partial U} F \cdot \nu \, dS(y)}$$

"Integrate by Parts" to put derivatives on Φ.

First, realize $\Delta_x f(x-y) = \Delta_y f(x-y)$.

Divergence Thm: $\displaystyle\int_{\mathbb{R}^n \setminus B(0,\varepsilon)} \operatorname{div}_y \left(\Phi(y) D_y f(x-y) \right) dy$



$$= \int_{\partial B(0,\varepsilon)} \Phi(y) D_y f(x-y) \cdot \vec{\nu} \, dS(y)$$

$$\int_{\mathbb{R}^n \setminus B(0,\varepsilon)} \operatorname{div}_y \left(\Phi(y) D_y f(x-y) \right) dy = \int_{\mathbb{R}^n \setminus B(0,\varepsilon)} D_y \Phi(y) \cdot D_y f(x-y)\, dy$$

$$+ \underbrace{\int_{\mathbb{R}^n \setminus B(0,\varepsilon)} \Phi(y) \Delta_y f(x-y)\, dy}_{= J_\varepsilon}$$

$$J_\varepsilon = \int_{\partial B(0,\varepsilon)} \Phi(y) D_y f(x-y) \cdot \vec{\nu}\, dS(y)$$

$$- \int_{\mathbb{R}^n \setminus B(0,\varepsilon)} D_y \Phi(y) \cdot D_y f(x-y)\, dy = K_\varepsilon + L_\varepsilon$$

$$|K_\varepsilon| = \left| \int_{\partial B(0,\varepsilon)} \Phi(y) Df(x-y) \cdot \vec{\nu} \, dS(y) \right|$$

As above, we have $|Df| \leq K_f$ everywhere.

$$\leq \int_{\partial B(0,\varepsilon)} |\Phi(y)| \, K_f \, dS(y)$$

$$\leq C_n K_f \int_{\partial B(0,\varepsilon)} \frac{1}{\varepsilon^{n-2}} \, dS(y)$$

because $|y| = \varepsilon$ on $\partial B(0,\varepsilon)$

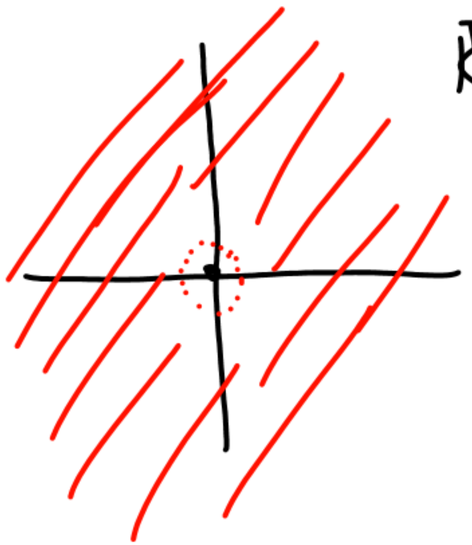
$$= C_n K_f \frac{1}{\varepsilon^{n-2}} \varepsilon^{n-1} = C_n K_f \varepsilon$$

$$L_\varepsilon = -\int_{\mathbb{R}^n \setminus B(0,\varepsilon)} D_y \Phi(y) \cdot D_y f(x-y)\, dy$$

$$\boxed{D_y |y| = \frac{y}{|y|}}$$

Integrate by parts again:

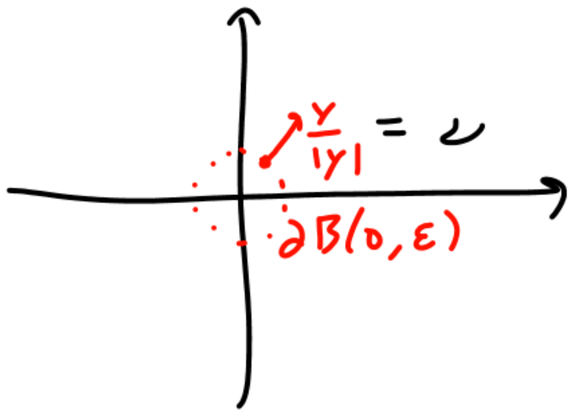
$$L_\varepsilon = \int_{\mathbb{R}^n \setminus B(0,\varepsilon)} \Delta_y \Phi(y)\, f(x-y)\, dy - \int_{\partial B(0,\varepsilon)} D_y \Phi(y) \cdot \vec{\nu}\, f(x-y)\, dS(y)$$



$$= 0 - \int_{\partial B(0,\varepsilon)} D_y \Phi(y) \cdot \vec{\nu}\, f(x-y)\, dS(y)$$

$$\Phi(y) = C_n \frac{1}{|y|^{n-2}}, \quad D_y \Phi(y) = C_n (2-n) |y|^{1-n} \frac{y}{|y|} = \frac{C_n(2-n)\, y}{|y|^n}$$

$$D_y \Phi(y) = c_n(2-n)\frac{y}{|y|^n}$$. What is $\vec{\nu}$?



$$\int_{\partial B(0,\varepsilon)} D_y \Phi(y) \cdot \vec{\nu}\, f(x-y)\, dS(y)$$

$$= \int_{\partial B(0,\varepsilon)} c_n(2-n)\frac{y}{|y|^n} \cdot \frac{y}{|y|}\, f(x-y)\, dS(y)$$

$$= \int_{\partial B(0,\varepsilon)} c_n(2-n)|y|^{1-n} f(x-y)\, dS(y) = \frac{c_n(2-n)}{\varepsilon^{n-1}} \int_{\partial B(0,\varepsilon)} f(x-y)\, dS(y)$$

$$= \frac{1}{n(n-2)\alpha(n)} \frac{(2-n)}{\varepsilon^{n-1}} \int_{\partial B(0,\varepsilon)} f(x-y)\, dS(y)$$

$$L_\varepsilon = \frac{-1}{n\alpha(n)\,\varepsilon^{n-1}} \int_{\partial B(0,\varepsilon)} f(x-y)\, dS(y)$$

$\alpha(n)\,\varepsilon^n$ = Vol of n-ball

$n\,\alpha(n)\,\varepsilon^{n-1}$ = Area of (n-1)-sphere

$$= -\fint_{\partial B(0,\varepsilon)} f(x-y)\, dS(y) \longrightarrow -f(x)$$

We've shown $\boxed{\Delta u(x) = -f(x)}$